

Lesson 6: Case Studies



6.0 Lesson Overview



Lesson Overview

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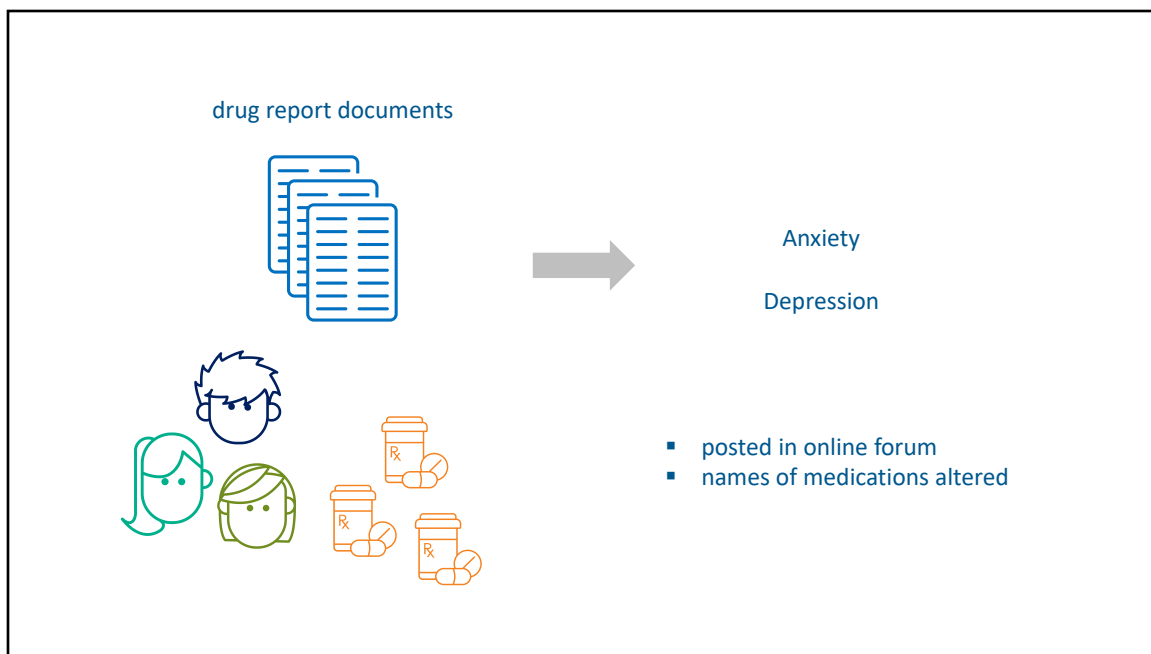
6.1 Information and Document Retrieval



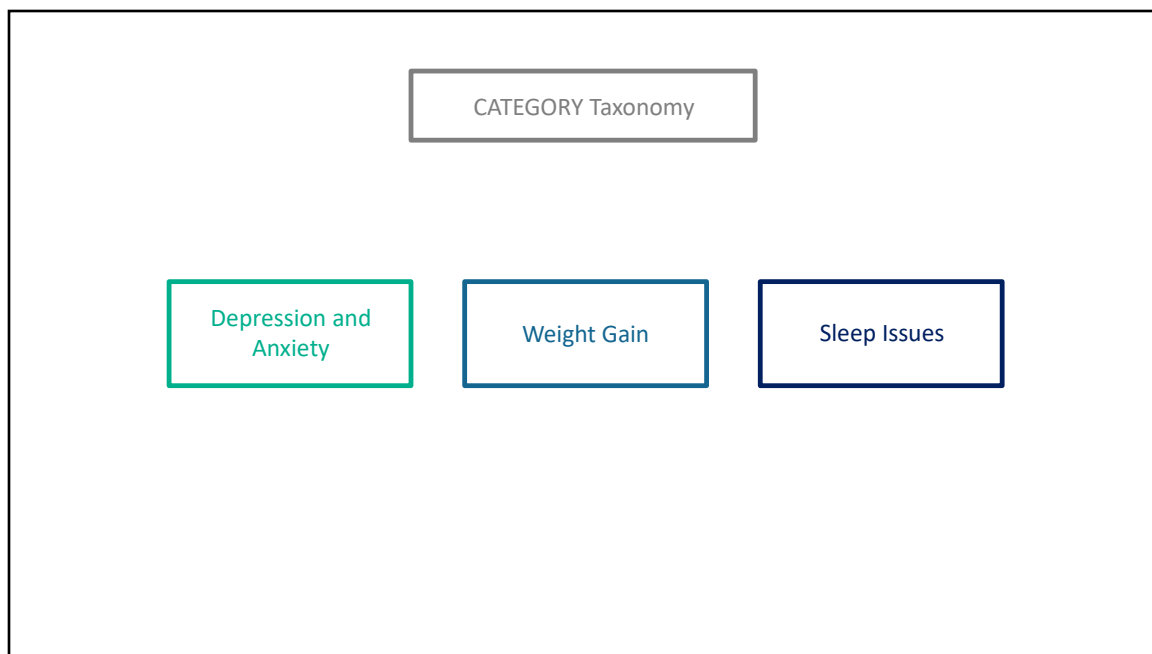
Drug Report Document Collection



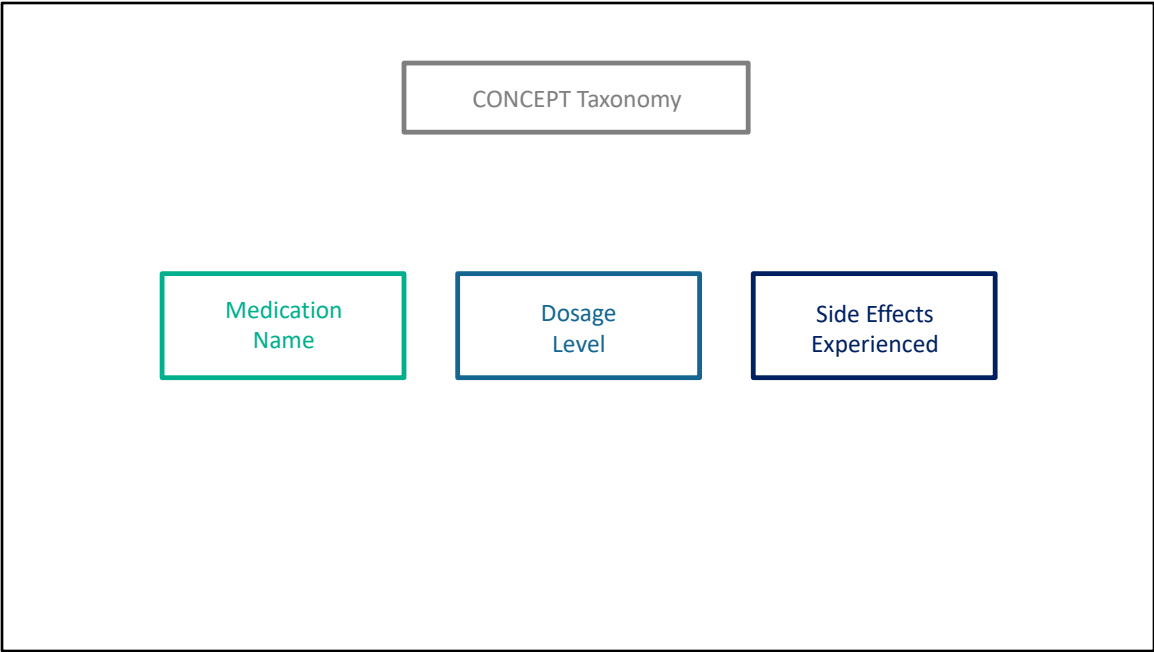
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In the first case study, we're going to retrieve information and documents about anxiety and depression from drug reports. The drug report documents contain comments from patients about prescription medications that are used for the treatment of depression. The comments were posted in an online forum, and the names of the medications that are referenced in the documents were altered. Rapid access to specific information can be enhanced through custom concepts, such as DOSAGE and PRESCRIPTION information.



Topics in the document collection include the following concerns: depression and anxiety, weight gain, and sleep issues. These will make up our Category Taxonomy.

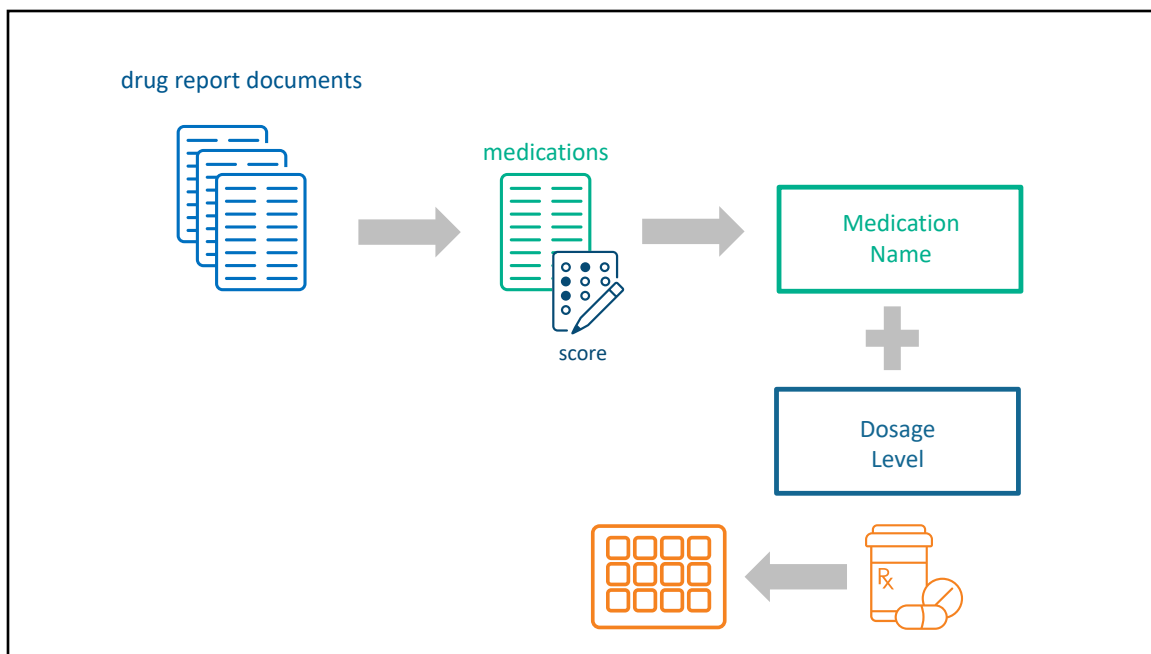


Some of the documents include the following key details: medication names, dosage levels, and side effects experienced. These will make up our Concept Taxonomy.

Concept	Examples
Medication	Abidal Escalan Cenerol
Dosage	20mg 60 MG 50 Mg
Prescription	Abidal 20mg 60 MG of Escalan 50 Mg Cenerol
SideEffects	Headache Blurred vision Nausea

Four custom concepts are defined in the Concepts node.

Medication with examples like Abidal, Escalan, Cenerol, etc. Dosage with examples like 20 mg, 60 MG, 50 Mg, etc. Prescription with examples like Abidal 20 mg, 60 MG of Escalan, 50 Mg Cenerol, etc. And SideEffects with examples like Headache, Blurred vision, Nausea, etc.



Our goal is to automatically extract information from the document collection. So for example, we'd like to obtain reports mentioning specific medications, so we have a Medication concept. When we score with respect to that concept, it's going to pull out all the medication names in the report. We will then combine that with dosage to get something called a Prescription. These concept rules will allow us to automatically parse the collection and create a table that has prescriptions in it. So we can immediately identify documents that mention a specific prescription, which would be a drug and a dosage.

Demo: Retrieving Information and Documents from Drug Reports, Part 1

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Demo: Retrieving Information and Documents from Drug Reports, Part 2

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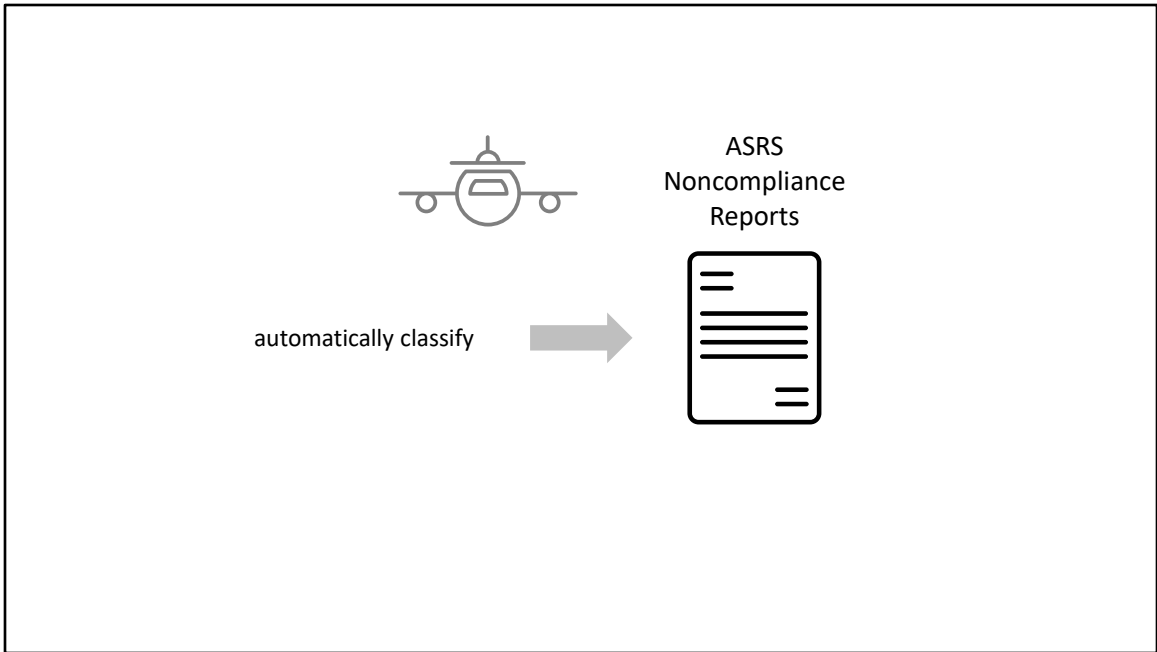
6.2 Automatic Categorization of ASRS Incident Reports



ASRS Noncompliance Reports



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We want to automatically classify ASRS procedure noncompliance reports. ASRS is the Aviation Safety Reporting System.



- fails to observe federal aviation regulation
- disregards an official notice
 - temporary runway closure
- fails to follow commands from air traffic controller
- violates temporary flight restriction
 - overflying major sporting event

The system is open to anyone who participates in aviation, including professional and private pilots, flight attendants, air traffic controllers, and airframe and powerplant mechanics. Procedure noncompliance occurs when a pilot fails to observe a federal aviation regulation, disregards an official notice such as a temporary runway closure, or fails to follow the commands given by an air traffic controller.

A common occurrence is to violate a temporary flight restriction, such as overflying a major outdoor sporting event. The ability to rapidly access specific safety information can facilitate timely action to prevent accidents.

Demo: Classifying ASRS Procedure Noncompliance Reports Automatically, Part 1

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Demo: Classifying ASRS Procedure Noncompliance Reports Automatically, Part 2

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Demo: Classifying ASRS Procedure Noncompliance Reports Automatically, Part 3

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Resources



Document Clustering



Resource

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Appendix A



Introduction

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Cheat Sheet Template: Concept Rules

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Cheat Sheet Template: Category Rules

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Appendix B



References

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